High School Internship Project

US Renewable Energy Transition

Presented by Aditya Brahmachary

OVERVIEW

Objectives

Data & Methodology

Renewable Trends & Pivot Year

Linear Regression

Key Methodology Caveat

State Report Card

Interactive Dashboard & 2014v2024 Comparison

Key Findings

Table of Contents

OBJECTIVES

Analysis behind 51 US States, 2001-2025 in 4 fuels: Coal, Natural Gas, Solar, Wind

Objective 01

Which states are winning the race to Net Zero? Which ones are falling behind the pack?

Objective 02

When did renewables overtake coal and when might each state fully transition?

Objective 03

How does the US look today compared to a decade ago?

Objective

INTRODUCTION

Source: EIA-923 Net Generation by State, 2001-2025
Scope: 50 States + DC → Coal, Natural Gas, Wind, Solar
Cleaning: Originally 256 columns used df.melt() to make the dataframe 6375 rows & 2 columns, stripped units, handled missing values, defined function to separate Fuel & State into 2 columns instead of being combined.
Reshape: Long forat EIA export into easy Multi-Index to understand.

The Entire Detailed Process can be seen in the Notebook!

		Generation	Fuel
Year	**State**		
2001	**Alabama**	125345.11300	all fuels (utility-scale)
	Alabama	72198.11400	coal
	Alabama	9612.68200	natural gas
	Alabama	NaN	wind
	Alabama	NaN	all solar
...	...	...	...
2025	**Wyoming**	46225.08347	all fuels (utility-scale)
	Wyoming	27024.11430	coal
	Wyoming	4594.53129	natural gas
	Wyoming	12525.58352	wind
	Wyoming	537.03917	all solar

6375 rows × 2 columns

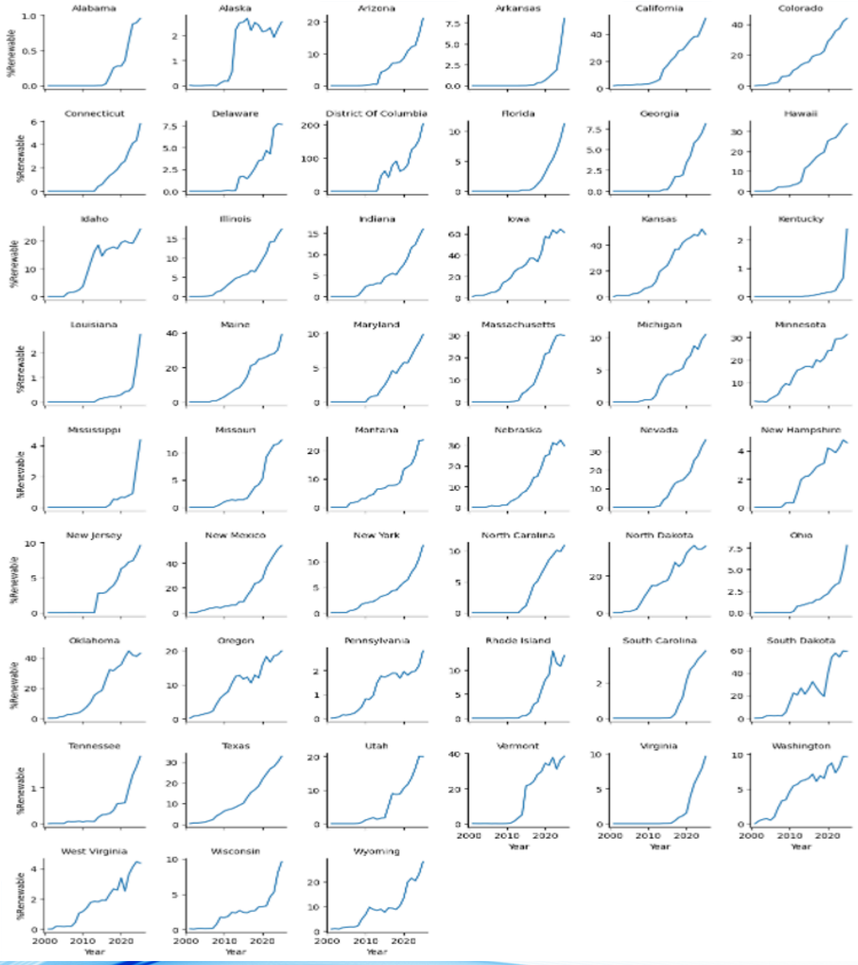
Data & Methodology

OVERALL TRENDS

22

states have not crossed the Pivot Year (where solar + wind > coal)

- California, Kansas, Vermont crossed early - avg: 2010-2016 ish
- Kentucky, West Virginia, etc have not crossed at all



Renewable Trends & Pivot Year

FORECASTING

Method: Linear Regression on each state's historical renewable, projected to the *Intersection Year* when wind + solar reaches 100% of generation

Note: Projections beyond 2100 were statistically useless and taken with a grain of salt.

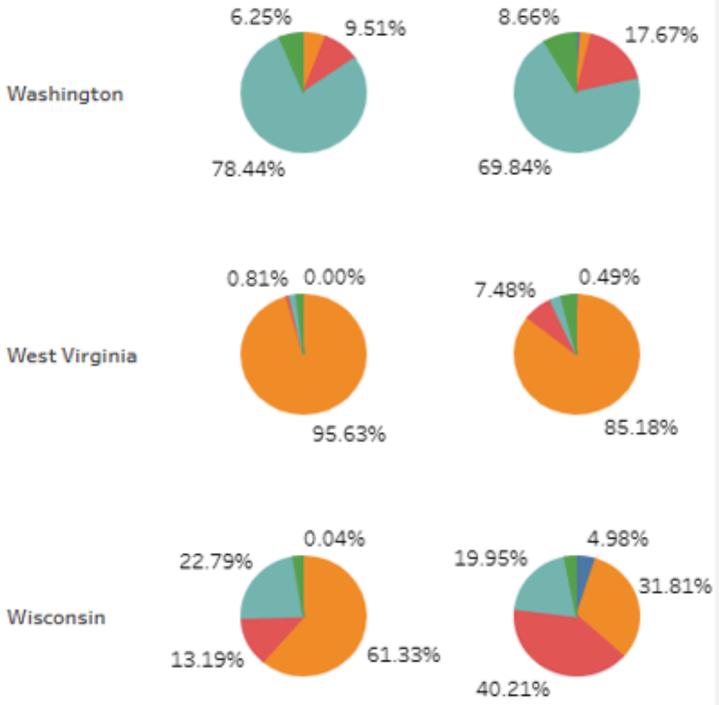
Top 5 States — Soonest Projected Net-Zero (Linear Regression)

State	Trend	lastest%	lastestYear	Intersection_Year	fg
Iowa	[2.971044404406448]	[61.4143034434759]	2025	2037.98725	Predicted
South Dakota	[2.6336432627936346]	[59.17663157485896]	2025	2040.500721	Predicted
Kansas	[2.5451857914632647]	[48.112501533054065]	2025	2045.386527	Predicted
New Mexico	[2.1366901046927036]	[54.02416211190385]	2025	2046.517317	Predicted
California	[2.058017832963995]	[51.659970831999935]	2025	2048.488635	Predicted

Linear Regression

MAJOR CRITIQUE

State	Trend	lastest%	lastestYear	Intersection_Year
Mississippi	[0.08771472144798395]	[4.36266118160302]	2025	3115.32255
Tennessee	[0.05896390027039084]	[1.8680874282049196]	2025	3689.27105
Louisiana	[0.05341265595248208]	[2.7304964451254237]	2025	3846.09467
Kentucky	[0.035512637261457036]	[2.3833670661681667]	2025	4773.78579
Alabama	[0.03403689615771339]	[0.9594382256324809]	2025	4934.80003



Key Methodology Caveat

Washington: 84% clean but ranks with a D grade. Why?

The model only counts winds + solar as renewable energy so states like Washington and Vermont that run on hydropower (a fuel type outside of the 4 analyzed) is ranked lower!

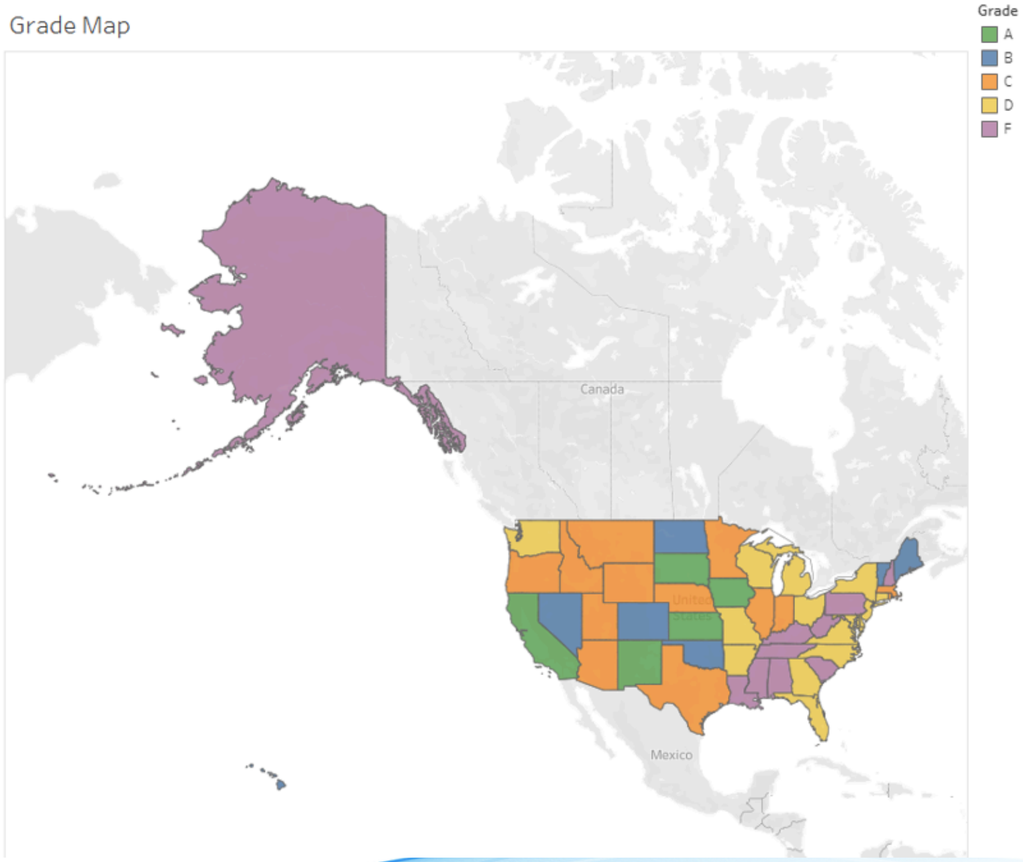
Definition of Linear Regression: expects continuous growth which laws might prevent. In addition, legislation might aim for an earlier target which is a factor outside of the data.

RESULT

Grade: 50% current renewable energy + 50% speed (trend)

Curve/Percentiles Note: absolutely relative to other states not national standard
(D/F are mostly SouthEast States)

90th Percentile → A
75th Percentile → B
50th Percentile → C
20th Percentile → D
0th Percentile → F

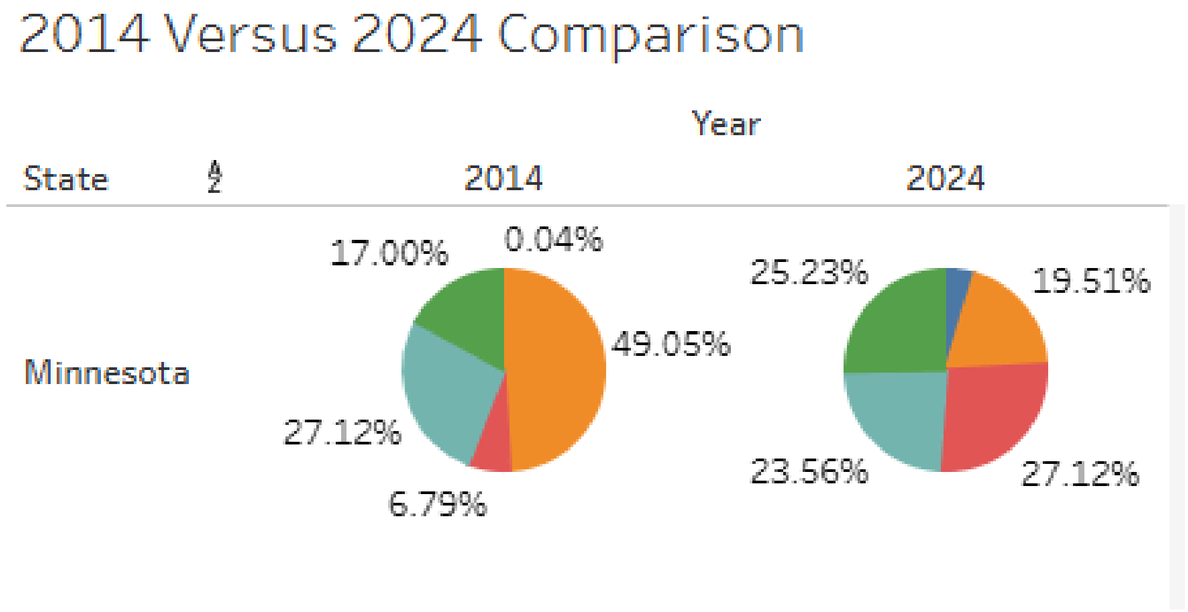


State Report Card

DECADE COMPARISON

https://10ay.online.tableau.com/t/adityabrahmachary2009-e944568002/views/StateofGrid2026Comparison/Dashboard1

You can check any state and see the INSANE difference between 2014 & 2024 fuel type usuage!



2014v2024 Comparison

ENDINGS

Leaders	Iowa, South Dakota, Kansas, New Mexico and California lead the nation in both renewable share and speed of adoption. They are the leaders to Net Zero!
Pivots	22 states show no clear coal-to-renewable crossover under current trends.
Blind spot	Hydro- and nuclear-heavy states are undercounted by a wind+solar-only definition. Linear Regression Definition. The data and code are excellent just the regression.
Recommendation	Track hydro/nuclear ("Other") as a separate category in future iterations for a fuller clean-energy picture.

Key Findings & Recommendations

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Thank You

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